

Reliability Engineering 101 for NSWC Corona

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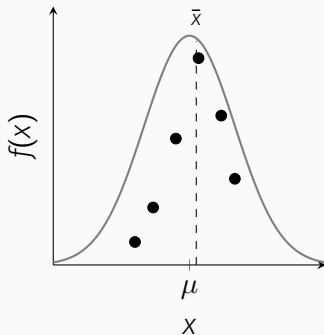
- 3.1 : Reliability of Repairable Systems
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Part I: Introduction and Mathematics

Module 1.2.3 : Estimation and Hypothesis Testing

Point estimation uses sample data to estimate unknown parameters of a probability distribution.

- Population parameters
 - Mean (μ)
 - Standard deviation (σ)
 - Failure rate (λ)
 - Weibull parameters (β, η)
- Sample estimates
 - Sample mean ($\bar{x}$)
 - Sample standard deviation (s)
 - Maximum likelihood estimators (MLEs)



Sample mean : Estimate the population mean from a finite sample.

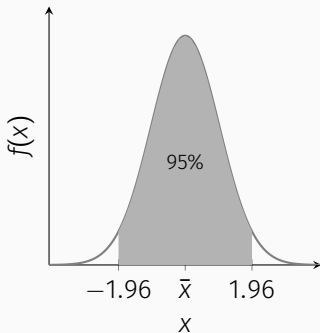
$$\bar{x} = \frac{1}{n} \sum_{i=1}^n x_i \quad (1)$$

Formulas for mean and variance are different for various PDFs.

Module 1.2.3 : Estimation and Hypothesis Testing

Confidence intervals quantify the uncertainty in an estimated parameter.

- Point estimates are never exact.
- Larger samples produce narrower intervals.
- Confidence levels are commonly
 - 90%
 - 95%
 - 99%



The graph represents a 95% confidence interval (CI) surrounding the estimated mean. CI for confidence level α :

$$\bar{x} \pm z_{\alpha/2} \frac{\sigma}{\sqrt{n}} \quad (2)$$

Module 1.2.3 : Estimation and Hypothesis Testing

Example : An electronic component is known to have an exponential failure distribution. A random sample of $n = 12$ components is tested, producing a total operating time of $S = \sum_{i=1}^{12} t_i = 8400$ hours. The estimated failure rate is

$$\hat{\lambda} = 1.43 \times 10^{-3} \text{hr}^{-1}. \quad (3)$$

The estimate for $\hat{\lambda}$ is derived from the maximum likelihood estimator (MLE, next page). Note that, for an exponential distribution, an exact CI for the failure rate is

$$\frac{\chi_{\alpha/2, 2n}^2}{2S} < \lambda < \frac{\chi_{1-\alpha/2, 2n}^2}{2S} \quad (4)$$

For a two-sided 95% confidence interval ($\alpha = 0.05$, $n = 12$), $\chi_{0.025, 24}^2 = 12.40$, and $\chi_{0.975, 24}^2 = 39.36$.

Question : (i) What is the 95% CI for the failure rate, and (ii) what is the corresponding confidence interval for the MTTF, if $MTTF = \frac{1}{\lambda}$?

- (i) Using Eq. ??,
 $7.38 \times 10^{-4} \text{hr}^{-1} < \lambda < 2.34 \times 10^{-3} \text{hr}^{-1}$.
(ii) Take the lower bound for λ as the inverse of the *upper* bound for MTTF, and vice versa: $427 \text{hr} < MTTF < 1355 \text{hr}$.

How do we obtain $\hat{\lambda}$?

Module 1.2.3 : Estimation and Hypothesis Testing

Maximum Likelihood Estimation (MLE): Suppose n independent failure times, $t_1, t_2, \dots, t_n$, are known to follow an exponential distribution with failure rate λ .

Question : How can the parameter λ be estimated from the observed data?

The likelihood function is

$$L(\lambda) = \prod_{i=1}^n \lambda e^{-\lambda t_i} = \lambda^n e^{-\lambda \sum t_i}. \quad (5)$$

Taking the natural logarithm,

$$\ln L(\lambda) = n \ln \lambda - \lambda \sum t_i \quad (6)$$

Differentiating and setting the result equal to zero,

$$\frac{d}{d\lambda} \ln L = \frac{n}{\lambda} - \sum t_i = 0 \quad (7)$$

Solving for λ gives the maximum likelihood estimator,

$$\hat{\lambda} = \frac{n}{\sum_{i=1}^n t_i} = \frac{1}{\bar{t}} \quad (8)$$

where $\bar{t}$ is the sample mean failure time.

Interpretation : The MLE chooses the value of λ that makes the observed failure times most likely to occur.

Module 1.2.3 : Estimation and Hypothesis Testing

Hypothesis testing determines whether an observed result is likely due to random variation or represents a real effect.

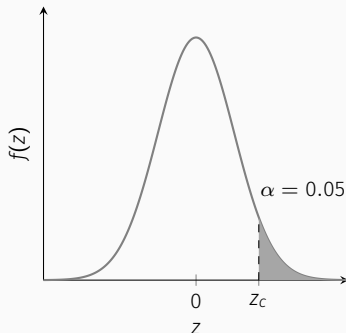
A hypothesis test begins by defining two competing hypotheses: (i) the **null hypothesis** (H_0), when no change or no effect exists, and the (ii) **alternative hypothesis** (H_1), when a real change or effect exists.

A test statistic (such as a z-score) is computed from the sample data and compared with the expected distribution assuming H_0 is true.

The **p-value** is the probability of observing a result at least as extreme as the measured value if the null hypothesis is true.

Decision Rule. Let α be a pre-determined threshold for the p-value. This means:

$p < \alpha$	$\Rightarrow$ Statistically significant
$p \geq \alpha$	$\Rightarrow$ Not statistically significant



Module 1.2.3 : Estimation and Hypothesis Testing

Example : A roller bearing has a lifetime that is normally distributed with $\mu = 6000\text{hr}$ and $\sigma = 450\text{hr}$. A sample of $n = 9$ bearings manufactured using a new lubricant has an average lifetime of $\bar{x} = 6400\text{hr}$.

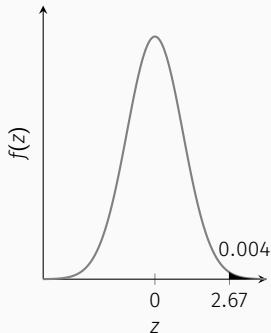
Question : Has the new lubricant produced a statistically significant increase in mean lifetime?

Assuming the *population* standard deviation, σ , is known, the error in the mean is $\sigma/\sqrt{n}$, the z-score is

$$z = \frac{|\mu - \bar{x}|}{\sigma/\sqrt{n}} \quad (9)$$

Substituting the data, $P(Z < 2.67) = 0.996$, from the standard normal table. The probability of observing a result this extreme by chance is therefore 0.004.

Result : Only a 0.4% chance exists that the increase in mean lifetime occurred randomly.



Since $0.4\% < 5\%$, the result is significant.

Module 1.2.3 : Estimation and Hypothesis Testing

Example : Suppose two samples of bearings from the previous example are tested. The first sample has a mean and standard deviation of 6000 hr and 450 hr ($n = 60$). The second sample has a mean and standard deviation of 6400 hr and 380 hr ($n = 9$).

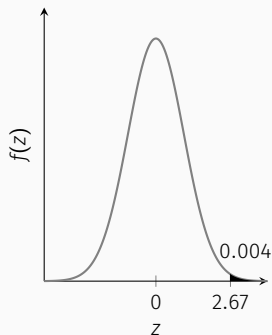
Question : Is the difference in mean values still statistically significant?

The standard deviation of the distribution of differences in the sample means is

$$S_{\bar{x}_1 - \bar{x}_2} = \frac{\sigma_1}{\sqrt{n_1}} + \frac{\sigma_2}{\sqrt{n_2}} \quad (10)$$

The z-score is the difference in sample means, divided by $S_{\bar{x}_1 - \bar{x}_2}$. Substituting the data, $z = 400/185 = 2.16$, and $P(Z < 2.16) = 0.985$, from the standard normal table. The probability of observing a result this extreme by chance is therefore 0.015.

Result : Only a 1.5% chance exists that the increase in mean lifetime occurred randomly.



Since $1.5\% < 5\%$, the result is significant.

Module 1.2.4 : Graphical and Tabular Methods

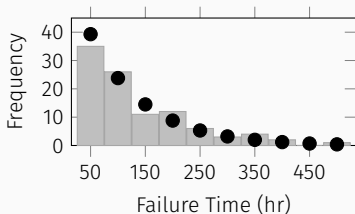
Graphical techniques are often useful to extract reliability parameters. Consider the following histogram of failure times measured from N units of the same electronic component.

Bin	Freq.	Exp. Freq.
0-100	35	39.3
100-200	26	23.8
200-300	11	14.5
300-400	12	8.8
400-500	6	5.3
500-600	3	3.2
600-700	4	2.0
700-800	2	1.2
800-900	0	0.7
900-1000	1	0.4

Table 1: Histogram of failure times (columns 1 and 2), alongside predicted failure times (column 3).

Assume the data follows the exponential distribution ($\lambda \approx 5 \times 10^{-3} \text{ hr}^{-1}$). The expected frequencies for events in each bin are:

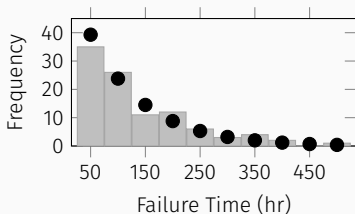
$$\text{Exp. Freq.} = N \int_a^b \lambda e^{-\lambda t} dt \quad (11)$$



Circles : expected frequencies. **Bars :** observed frequencies.

Module 1.2.5 : Goodness-of-Fit Tests

Goodness-of-fit tests assess the agreement between predicted and observed distributions.



Let o_i represent the observations (gray bars), and e_i represent the predictions (black circles). The *chi-square* test compares the following test statistic to the Chi-square PDF:

$$\chi^2 = \sum_{i=1}^N \frac{(o_i - e_i)^2}{e_i} \quad (12)$$

Basic steps :

- Choose distribution and confidence level (CL) $1 - \alpha$
- Let N be the number of **bins**
- Let n be the number of distribution parameters
- Calculate the sum in Eq. ??
- If the result is greater than $\chi^2_{1-\alpha}(N - n - 1)$, reject the fit.
- Else, accept the fit at CL $1 - \alpha$.

Module 1.2.5 : Goodness-of-Fit Tests

- Select exponential model, with $\lambda = 5 \times 10^{-3} \text{ hr}^{-1}$.
- $N = 10, n = 1$.
- Calculate χ^2 sum: For the exponential failure model, $N - n - 1 = 8$. The measured and expected frequencies give:

$$\chi^2 = \sum_{i=1}^{10} \frac{(o_i - e_i)^2}{e_i} = 6.92 \quad (13)$$

- Compare to theoretical distribution:

$$1 - \alpha = 0.95 \quad (14)$$

$$\chi_{0.95}^2(8) = 15.51 \quad (15)$$

$$6.92 < 15.51, \text{ accept model} \quad (16)$$

Microsoft Excel, $\chi_{1-\alpha}^2(N - m - 1)$:

=CHISQ.INV.RT(1-alpha, N-n-1)

Figure ?? : Algorithms can also minimize the *reduced chi-square*:

$$\chi_r^2 = \frac{1}{N-n} \sum_{i=1}^N \frac{(f_i - f(x_i))^2}{\sigma_i^2}.$$

Fitting the *reduced chi-squared* to un-binned data is common in the physical sciences.

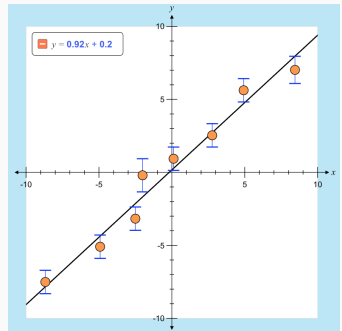


Figure 1

Module 1.2.5, 1.2.6 : Goodness-of-Fit Tests, Linear Regression

Reduced chi-square compares several candidate models while accounting for the number of fitted parameters. For example, consider the model table below.

Model	χ^2	χ_r^2	Result
Exponential	16.8	2.40	Poor fit
Lognormal	10.8	1.54	Acceptable
Weibull ($\beta = 1.35$)	7.1	1.01	Best fit

For all three models, the degrees of freedom (dof) are $N - n - 1 = 7$. The reduced chi-square is computed as

$$\chi_r^2 = \frac{\chi^2}{\text{dof}}. \quad (17)$$

For the Weibull model, $\chi_r^2 = 7.1/7 = 1.01$. The interpretation of χ_r^2 is as follows:

- $\chi_r^2 \gg 1$: model poorly explains the data.
- $\chi_r^2 \approx 1$: model agrees with observations.
- $\chi_r^2 \ll 1$: over-fitting the data.

The Weibull model has the smallest χ_r^2 , so $\beta = 1.35$ is selected.

Note: when tuning β to minimize χ_r^2

- $\beta > 1$ indicates failure rate is increasing with time.
- $\beta < 1$ indicates failure rate is decreasing with time.

Linearized data $\Rightarrow$ regression techniques:

